

Week	Monday	Wednesday	Friday
Week 1	01/13/25 Course overview and early works of sci-fi Critique of the first sci-fi film Imagining human spaceflight in 1902 Science knowledge survey (25 pts)	01/15/25 Reference frames, acceleration and artificial gravity	01/17/25 What is the nature of space and time? Classical physics and Newton's laws Quiz 1: Intro material (20 pts)
Week 2	01/20/25 No Class: Martin Luther King Jr Day	01/22/25 Newton's laws continued Einstein and modern physics	01/24/25 Einstein and modern physics Relativistic spacetime Problem 1: acceleration (25 pts)
Week 3	01/27/25 Einstein and modern physics Special relativity and time travel	01/29/25 General relativity Black holes and gravitational waves	01/31/25 No Class: Spacetime Team Project Problem 2: time dilation (25 pts)
Week 4	02/03/25 General relativity continued Gravitational waves Exploration Paper 1: Space & Time (25 pts)	02/05/25 The warp drive and "Future physics": Quantum gravity Spacetime Team Project (25 pts)	02/05/25 "Future physics": Quantum gravity, worm holes, and the Multiverse Hypothesis
Week 5	02/10/25 What is the Universe made of? The standard model of particle physics Problem 3: General relativity (25 pts)	02/12/25 Dark matter and dark energy Quiz 2: Space and Time (85 pts)	02/14/25 Nuclei, atoms, and gases
Week 6	02/17/25 Liquids, solids, phase transitions	02/19/25 Optical properties; transparent materials	02/21/25 Metamaterials and invisibility cloaks Problem 4: matter & materials (25 pts)
Week 7	02/24/25 Chemical energy Midterm Film Critique (50 pts)	02/26/25 Nuclear weapons; matter & antimatter	02/28/25 Can a machine become self aware? Information storage & data processing Exploration Paper 2: Matter & Energy (25 pts)

Week 8	03/03/25 No Class: Spring Break	03/05/25 No Class: Spring Break	03/07/25 No Class: Spring Break
Week 9	03/10/25 Intelligent robots	03/12/25 The Turing Test Quiz 3: Matter and Energy (85 pts)	03/14/25 Asimov's Laws and robot behavior
Week 10	03/17/25 Artificial consciousness & robot ethics	03/19/25 Are we alone in the Universe? Conditions necessary for life to exist Exploration Paper 3: AI (25 pts)	03/21/25 Crewed/un-crewed spaceflight Finding habitable planets
Week 11	03/24/25 Finding habitable planets continued Quiz 4: AI (50 pts)	03/26/25 The science behind the SETI project	03/28/25 No class: SETI Team Project Exploration Paper 4: SETI (50 pts)
Week 12	03/31/25 What does it mean to be human? Insight from Frankenstein Medical technology Quiz 5: Extraterrestrial Intelligence (30 pts)	04/02/25 Cell structure and radiation damage DNA, genome sequencing & gene editing SETI Team Project Due (25 pts)	04/04/25 No Class: Spring Carnival
Week 13	04/07/25 Human teleportation	04/09/25 What can we learn from an android about what it means to be human?	04/11/25 How can we solve our problems Public perception of science The scientific method Exploration Paper 5: Being Human (25 pts)
Week 14	04/14/25 Misplaced faith in science Misunderstanding of science Quiz 6: Being Human (50 pts)	04/16/25 Misuse of science in advertising Final Film Critique (50 pts)	04/18/25 What lies ahead? Examples of sci-fi predicting the future How sci-fi storytelling changes with time
Week 15	04/21/25 No class	04/23/25 In class critiques Quiz 7: Problem solving (30 pts)	04/25/25 In class critiques Last day of class – closing discussion